

Towards Understanding Financial Decision-Making in Equity Markets Among German and Indian Participants: Survey-based Findings

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Abstract

In many economies, individuals increasingly rely on equity market participation for long-term wealth accumulation and pension provision. While broader access to capital markets offers significant potential for improving financial well-being, it also exposes investors to market volatility and the risk of financial losses. Such volatility can trigger emotional responses that lead to suboptimal decisions, including panic selling. This paper investigates financial decision-making in equity markets with a focus on investors' behavioral responses to risk and uncertainty. We conduct a web-based survey ($N = 42$) among participants in Germany and India. The preliminary results indicate promising directions for future research, including investor behavior modeling using physiological indicators and the development of personalized financial product recommendation algorithms based on individual risk profiles.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → *Computing and business*; User characteristics.

Keywords

financial decision making, behavioral finance, survey-based analysis, physiological response

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1 Introduction

Because high-stakes individual financial decisions are infrequent, high-impact, and difficult to reverse, integrating behavioral finance insights into the design of investment advice and decision-support systems is receiving increasing attention in HCI research [3, 4,

10, 12, 14]. We explore investor behavior and equity-related financial decision-making by jointly examining self-reported risk attitudes [13], physiological responses to market downturn scenarios [8] and preferences over structured product payoff profiles [11]. The choice of payoff profile is related to different types of value functions in prospect theory (e.g. risk-aversion over gains, risk-seeking over losses). By integrating these three perspectives, we seek to gain new insights into how risk is perceived, experienced, and ultimately reflected in payoff choices. The analysis also allows us to investigate whether individuals' subjective risk perceptions and affective reactions align with their preferences for concrete financial decisions.

The preliminary analysis presented here is based on an initial sample of respondents ($N=42$) with a finance background, primarily from India and Germany, and is intended as a first step toward assessing the consistency and interpretability of the proposed measures. Despite its exploratory nature, the results already suggest meaningful patterns in how risk is perceived and translated into payoff preferences, as well as interesting indications of both inter-cultural commonalities and differences.

2 Behavioral Models and Financial Product Choice

Integrating concepts of behavioral finance in investment advice is an important development in financial economics. This includes identifying cognitive behavioral biases (e.g. hindsight bias, conservatism) and emotional behavioral biases (e.g. loss aversion bias, overconfidence, herding) and adapting investment advice according to the biases identified and investors' wealth levels [17, 18]. Conducting surveys on financial behaviour and risk attitudes involving emotional arousal effects provide new insights into investment decision-making, e.g. [6].

As part of the survey, participants are presented with four risky payoff profiles corresponding to structured products commonly offered in the German retail market as so-called *certificates* (*Zertifikate*). Hens and Rieger [11] show that investors whose preferences are described by classical expected utility theory would typically prefer direct investments in the underlying asset, whereas non-linear payoff profiles can be rationalised when preferences follow (cumulative) prospect theory. Eliciting preferences over these payoff profiles therefore allows us to relate implicit payoff-based risk preferences to self-reported risk attitudes.

In classical utility theory, investor preferences are represented by a twice continuously differentiable, increasing and concave utility



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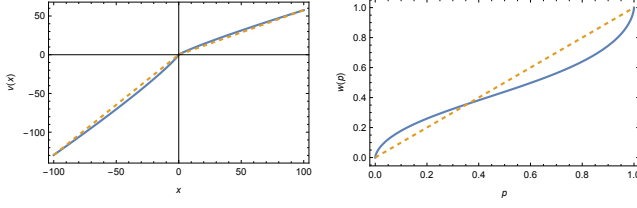


Figure 1: Left: value function v with parameters specified in Tversky and Kahneman [21]; comparing with the linear dotted line shows that v is convex in losses and concave in gains. Right: Probability weighting function w with parameter specified in Barberis [1]; comparing with the linear dotted line shows that small probabilities are overweighted.

function $u : \mathbb{R} \rightarrow \mathbb{R}$. Given a payoff function $y : \mathbb{R} \rightarrow \mathbb{R}$ on a risky security, expected utility is $U(y) = \int_{\mathbb{R}} u(y(x)) p(x) dx$, subject to the budget constraint $\int_{\mathbb{R}} y(x) q(x) dx = R$, where p is the probability density of realisations of the risky security under the real-world measure $\mathbb{P}$ and q is the probability density of the pricing measure $\mathbb{Q}$. For standard CRRA utility (constant relative risk aversion), expected utility is maximised by near-linear payoffs, implying limited demand for structured products [11]. This motivates behavioral extensions.

Under cumulative prospect theory [1, 21], outcomes are evaluated relative to a reference point using a value function v and probability weighting. For a gamble $\{(x_i, p_i)\}$, overall valuation is given by

$$\sum_i \pi_i v(x_i), \quad \text{where } v(x) = \begin{cases} x^\alpha, & x \geq 0, \\ -\lambda(-x)^\beta, & x < 0, \end{cases}$$

is the value function and decision weights π_i are derived from a probability weighting function $w(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{1/\delta}}$. The left graph in Figure 1 shows the value function proposed by Tversky and Kahneman [21] with $\alpha = \beta = 0.88$ and $\lambda = 2.25$. The right graph in Figure 1 shows the probability weighting function with $\delta = 0.65$ as specified in Barberis [1]. This framework captures reference dependence, loss aversion, diminishing sensitivity, and probability weighting, providing a behavioral explanation for demand for non-linear payoff structures in retail markets.

3 Web-based Survey: Tools and Method

We designed a web-based survey¹ (approved by our institute's *Ethics Committee*) covering the following aspects – demographic details, financial profile details, risk perception, and financial decision making. We describe them below.

3.1 Survey Questionnaires

3.1.1 Demographic and Socioeconomic Details. The survey begins with questions on gender, age group, and country of residence, which provide essential demographic segmentation. Following this, participants are asked about their highest education level, which

serves as an indicator of financial literacy potential, and their gross annual income quintile (country-wise reference was mentioned in the survey), categorized relative to their country of residence.

3.1.2 Financial Profile Details. This section collects detailed information on participants' asset holdings and investment history. Questions cover the current equity portfolio value, investment duration in the equity market. The section also captures self-assessed risk profiles (conservative, moderate, aggressive, unsure), frequency of portfolio monitoring, which reflects the level of engagement and activity in financial decision-making.

3.1.3 Risk Perception and Attitudes toward Investment Volatility. This part explores participants' psychological responses to financial risk. The risk perception questions are broadly guided by the following well-known financial profiling questionnaires - Quirion's Questionnaire [19], Cominvest's Questionnaire [5], Growney's Questionnaire [9]. This section includes Likert-scale questions on risk aversion, such as the extent to which they are weighed down by potential permanent losses, their prioritization of investment security, reluctance to take risks, and nervousness about small losses. On the other hand, it also assesses risk-seeking tendencies, tolerance for annual fluctuations in investment value, and thresholds for what constitutes a 'significant loss' within a quarter. Subsequent questions probe into immediate emotional reactions for a significant loss-ranging from panic and anxiety to resilience and calmness. Finally, the survey asks how participants typically deal with significant losses, with multiple options such as consulting financial advisors (FA), following expert opinions, or capitalizing on opportunities to buy more equities.

3.1.4 Financial Decision-Making. This part presents a choice-based experiment involving four structured investment products, whose payoffs are shown in Figure 2. From left to right, the products are: An investor with a concave risk profile would favor the *Capital protection* certificate. An investor with the value function determined by Tversky and Kahneman [21] favours the *Airbag* certificate. An investor with strong diminishing sensitivity (strongly convex in losses, strongly concave in profits) prefers the *Twin win* certificate. A risk-neutral investor prefers the *Sprint* certificate. Each of the four structured products have the same initial price. It is assumed that the price process of the risky asset follows a Black-Scholes model (Geometric Brownian motion) with drift $\mu = 0.09$ and volatility $\sigma = 0.2$. The risk-neutral rate is $r = 0.04$. The notional amount of the underlying is 1 and expiry of the structured product is in $T = 1$.

Each product's payoff profile is visually depicted and explained, and participants are asked to select their preferred option for a one-year investment scenario. This task aims to capture realistic decision-making under uncertainty, moving beyond self-reported attitudes to actual choices. The questionnaire concludes with an open-ended comment box, allowing participants to provide feedback on question clarity, survey design, and contextual factors such as income security, and life-stage influences on investment behavior.

3.2 Survey Method

The survey was performed via a Google form. The participants were emailed the survey link and were asked to fill out the survey.

¹We present the survey questionnaires in the supplementary material.

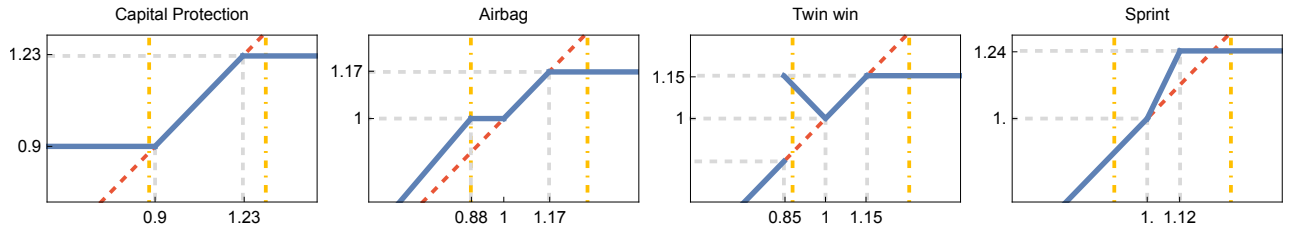


Figure 2: Payoff profiles from structured product offered to subjects. The yellow dashed lines show the range where the payoff is expected to be realised with approximately 2/3 probability (66%).

Income quintile	India	Germany
First quintile	7.1%	21.4%
Second quintile	14.3%	14.3%
Third quintile	3.6%	–
Fourth quintile	7.1%	7.1%
Fifth quintile	67.9%	57.1%

Table 1: Population distribution across income quintiles

Equity portfolio value (USD)	India	Germany
No Equity	46.4%	7.1%
<5K	17.9%	21.4%
5K–10K	21.4%	7.1%
10K–30K	3.6%	14.3%
30K–50K	10.7%	–
50K–200K	–	14.3%
200K–400K	–	7.1%
>400K	–	28.6%

Table 2: Population distribution in terms of current equity portfolio value

Equity investment experience	India	Germany
Never Invested	25.0%	14.3%
<1 year	17.9%	–
1–5 years	46.4%	35.7%
5–10 years	7.1%	14.3%
10–20 years	3.6%	7.1%
>20 years	0.0%	28.6%

Table 3: Population distribution in terms of equity investment experience

The instructions and objective of the survey were noted in survey form itself. Based on our preliminary demo runs, we anticipated the survey to take 10–12 minutes. We performed the survey among 42 participants (32 M, 9 F, 1 participant did not disclose the gender) in our universities. There are 45.2%, 16.7%, 4.7%, 9.5%, 7.1%, 4.7%, 11.9% participants in the 18–25 years, 26–30 years, 31–35 years, 36–40 years, 41–45 years, 46–50 years, and 50+ years respectively. We have 28 participants from India (one was in Singapore during the survey), 14 participants from Germany. The participants include current postgraduate students, PhD scholars, employees, and recent graduates.

4 Survey Findings

4.1 Financial Profile Overview

We present the distribution of the survey participants in terms of gross annual income, equity portfolio value, and exposure to the equity market in Tables 1, 2 and 3 respectively. The data reveals a large number of the participants belong to the highest income brackets in their respective countries (Table 1). The chi-square test does not reveal a significant difference in the gross income of the participants ($\chi^2(4, N = 42) = 2.27, p = .69$).

The survey also reveals almost half of the participants from India do not have any equity asset, whereas $\approx 35\%$ of German participants have a significant sum in their equity portfolio (Table 2). The difference in equity portfolio values are found to be significantly different among the participants of these countries ($\chi^2(8, N = 42) = 23.33, p = .003$). This could be attributed to the lesser familiarity with the equity instruments among Indian participants and because of the absolute difference in the wages of these two countries.

We also note that a large number of participants from Germany have a significant (more than 10 years) years of equity experience (Table 3). The equity investment experience is also found to be significantly different ($\chi^2(5, N = 42) = 12., p = .035$). This could be attributed to the age distribution of the survey participants as we have relatively younger participants from India. Nonetheless, there are 25% participants from India, who have no equity investment experience.

4.2 Risk Profile

We also analyze the risk profile of the survey participants. We use three factors for this – personalized risk categorization, volatility tolerance and acceptable draw-down in a quarter. We present the findings in Table 4. In terms of the self-reported risk profile, Indian participants are found to be more risk-averse (conservative), while participants from Germany are found to be more aggressive (28.6%). However, we do not observe a significant difference in the self-reported risk preferences of the participants ($\chi^2(3, N = 42) = 3.6, p = .31$).

We also noted a higher level of volatility tolerance among the participants from Germany. A large number of German participants ($\approx 72\%$) are found to be comfortable with at least 15% volatility, whereas 32% and 39% Indian counterparts are found to be comfortable with 5% and 10% volatility respectively. The chi-square test also reveals a significant difference among the participants ($\chi^2(3, N = 42) = 7.02, p = .071$) in terms of volatility tolerance.

In the same vein, 42.9% and 28.6% participants from Germany are comfortable with larger amount of quarterly loss (10–20% and 20–30% respectively). However, we do not observe a significant difference in this case ($\chi^2(4, N = 42) = 3.75, p = .44$). We believe one

	Risk profile				Volatility				Acceptable loss (in a quarter)				
	Not sure	Conserv.	Moderate	Aggressive	±5%	±10%	±15%	>15%	1–3%	3–5%	5–10%	10–20%	20–30%
India	14.3%	39.3%	32.1%	14.3%	32.1%	39.3%	14.3%	14.3%	14.3%	10.7%	32.1%	32.1%	10.7%
Germany	–	28.6%	42.9%	28.6%	14.3%	14.3%	35.7%	35.7%	7.1%	7.1%	14.3%	42.9%	28.6%

Table 4: Population distribution in terms of self-reported risk preferences, volatility tolerance, and loss levels perceived as significant levels. Participants from Germany have comparatively higher Aggressive risk preference, comfortable with larger volatility, and higher amount of quarterly loss.

	Behavioural reactions		Common actions			
	Stay calm	Emotional reaction	No action	Contact FA*	Seek expert opinion	Invest more
India	51.8%	48.1%	39.3%	3.6%	42.9%	14.3%
Germany	50.0%	50.0%	50.0%	14.3%	7.1%	28.6%

Table 5: Distribution of several behavioural reactions and common actions taken by the participants in case of a significant loss in the portfolio. *FA implies financial advisor.

	India	Germany
Capital protection	22.2%	29.6%
Airbag	18.5%	3.7%
Twin win	22.2%	11.11%
Sprint	11.1%	3.7%
I am not sure	25.9%	3.7%

Table 6: Product choice distribution by product type and country

	Conservative	Moderate	Aggressive	Not sure
Capital protection	33.3%	13.3%	75%	25%
Airbag	13.3%	20%	0%	25%
Twin win	13.3%	46.7%	12.5%	0%
Sprint	13.3%	6.7%	0%	25%
I am not sure	26.7%	13.3%	12.5%	25%

Table 7: Product choice distribution by product type and risk profile

of the primary reasons could be that among the survey participants, German participants are having comparatively longer duration of equity exposure (Table 3) and therefore, they are familiar with the volatility in the equity market and adjusted their risk profiles.

4.3 Behavioral Reactions and Decision Making

This section examines participants' behavioral reactions and decision-making in response to significant portfolio losses. Comparative results are reported in Table 5. Participants from both countries show similar response patterns: approximately 50% report remaining calm during substantial losses, while another 50% experience emotional reactions. These include stress, anxiety, reduced concentration, and, in some cases, changes in facial expressions and physiological indicators such as increased heart rate.

We also checked the common actions taken by the participants in case of a significant loss. We observe a noticeable difference in the number of participants seeking expert opinion and contacting the personal financial advisors between Germany and India. However, in both the cases, there are sizable number of participants, who do not take any action. In both the cases, some participants mentioned that they use such a volatility as an opportunity to invest more.

4.4 Financial Product Preference

We present the financial product preferences (Capital protection, Airbag, Twin win, Sprint) among the participants in Table 6. We compare the product preferences between the survey participants of

these countries using a χ^2 test, which does not reveal a significant difference between the product preferences ($\chi^2(4, N = 42) = 6.06$, $p = .19$). We also looked into the relationship between product selection and self-reported risk preferences (Conservative, moderate, aggressive) in Table 7. In this case also, we did not observe a significant association for selected products and risk profile ($\chi^2(12, N = 42) = 16.33$, $p = .177$).

4.5 Key Takeaways

To summarize, first, participants from the two countries exhibit significantly different levels of equity exposure, despite no significant differences in gross income (measured by relative income quintiles). Second, while self-reported risk preferences and experienced quarterly losses do not differ significantly across countries, participants' comfort with market volatility differs markedly. Third, no significant cross-country differences are observed in product preferences. Finally, product preferences do not vary significantly across risk profiles.

5 Discussion and Future Research Agenda

Despite the small sample size, homogeneous study population (university students and recent graduates) and focus on two countries, the results point to several avenues for future research. **First**, the lack of differentiation in product selection across risk profiles may create mismatches that increase investor stress, highlighting the need for improved recommendation strategies aligned with

individual risk preferences [2, 15]. **Second**, nearly half of participants in both countries reported emotional reactions during severe market drawdowns (Table 5), suggesting that physiological indicators—alongside traditional characteristics such as personality [7]—could enhance investor profiling. Given the widespread use of wearable devices [16, 20, 22], such data may be readily available for personalized financial advice to have better financial outcome and user experience. **Finally**, cross-country differences motivate further studies with larger and more diverse samples (beyond university students and with different levels of financial literacy) to examine decision-making across risk profiles.

6 Conclusion

In this paper, we conducted a web-based survey to investigate the financial decision making and behavioral actions of investors from India and Germany. The preliminary analysis from the survey reveals that despite having similar income profile (in terms of relative position in the respective countries), participants have different equity exposure, different risk profiles and different kinds of emotional reactions during significant draw-down. More interestingly, the participants selected financial products that do not corroborate with risk profiles. In summary, this study works as a precursor for future works in terms of better investor profiling (accounting for behavioral biases) and optimized financial product recommendation.

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